

The effects of various auditory takeover requests: A simulated driving study considering the modality of non-driving-related tasks

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ABSTRACT

With the era of automated driving approaching, designing an effective auditory takeover request (TOR) is critical to ensure automated driving safety. The present study investigated the effects of speech-based (speech and spearcon) and non-speech-based (earcon and auditory icon) TORs on takeover performance and subjective preferences. The potential impact of the non-driving-related task (NDRT) modality on auditory TORs was considered. Thirty-two participants were recruited in the present study and assigned to two groups, with one group performing the visual N-back task and another performing the auditory N-back task during automated driving. They were required to complete four simulated driving blocks corresponding to four auditory TOR types. The earcon TOR was found to be the most suitable for alerting drivers to return to the control loop because of its advantageous takeover time, lane change time, and minimum time to collision. Although participants preferred the speech TOR, it led to relatively poor takeover performance. In addition, the auditory NDRT was found to have a detrimental impact on auditory TORs. When drivers were engaged in the auditory NDRT, the takeover time and lane change time advantages of earcon TORs no longer existed. These findings highlight the importance of considering the influence of auditory NDRTs when designing an auditory takeover interface. The present study also has some practical implications for researchers and designers when designing an auditory takeover system in automated vehicles.

1. Introduction

1.1. Conditional automated driving and auditory takeover requests

Automated driving is currently a popular topic, and it is considered an effective solution for alleviating traffic congestion, improving the driving experience, and saving on labor costs (Hussain and Zeadally, 2019). SAE International (2021) categorized driving automation from Level 0 (no driving automation) to Level 5 (full driving automation). Partially automated driving (L2) has entered the vehicle marketplace, and conditional automated driving (L3) is expected to be commercially deployed soon. When the L3 automated driving system is activated, the drivers no longer need to monitor or control the system. Instead, they can engage in non-driving-related tasks (NDRTs), such as playing games, watching movies, and listening to music. Nevertheless, situations still

exist in which the automated driving system reaches a boundary that it cannot address (e.g., an upcoming collision). In these situations, the drivers must take over control of the automated vehicle to ensure their driving safety. Designing an effective takeover request (TOR) is, therefore, critical in ensuring automated driving safety.

The modality is one of the most critical factors influencing the effectiveness of TORs. Many studies in the human factors domain (e.g., Geitner et al., 2019; Petermeijer et al. 2017a; Yoon, Kim, & Ji, 2019; see McDonald et al., 2019 for a review) have explored the effect of the TOR modality on takeover performance. There are usually three modalities that researchers use to present TORs, namely visual, auditory, and tactile. It has been found that drivers often miss visual TORs because their heads and eyes are oriented toward their NDRT (Yoon et al., 2019). Compared with visual TORs, auditory and tactile TORs can easily attract drivers' attention and shorten their response times (Chai et al., 2022; Shi

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et al., 2022; Yoon et al., 2019). However, tactile TORs may lead to driver annoyance and poor experiences (Petermeijer et al., 2016). Contrastingly, auditory TORs are more user-friendly and preferred by people (Bazilinskyy et al., 2018), so they are increasingly being recommended and applied in current automated vehicles (e.g., Tesla, 2020).

1.2. The potential of different types of auditory takeover requests

1.2.1. Speech-based warnings

Auditory warnings in vehicles can be mainly categorized into speech-based and non-speech-based warnings (Noyes et al., 2006). Speech-based warnings mainly include speech and spearcon warnings (Song et al., 2022). Speech warnings contain specific semantic information that alerts drivers of driving-related events. People often rely on speech to communicate with each other. Therefore, speech is a common way for machines to convey information to users. Additionally, with the development of technology, it is increasingly convenient to generate speech with the text-to-speech tool. Speech warnings are often used to alert drivers of potential hazards in a manually driven vehicle (e.g., Chang et al., 2009; Yan et al., 2015; Zhang et al., 2019b). In an automated vehicle, speech is also widely used as a TOR to alert drivers to return to the control loop (e.g., Forster et al., 2017; Ma et al., 2021; Wang, Zhang, & Zhou, 2022). Speech warnings or TORs are generally easy to understand and learn (Noyes et al., 2006). However, drivers need to hear the complete content of the speech warning to understand its meaning due to speech being linear output. Therefore, speech may result in drivers responding more slowly (Walker et al., 2013).

Based on speech, researchers have developed spearcons (Walker et al., 2006) by speeding up speech by 40–50 % while maintaining the pitch. Previous studies have demonstrated that spearcons can lead to a shorter response time than speech without sacrificing response accuracy (Sabic and Chen, 2016; Walker et al., 2013). In addition, Dingler et al. (2008) claimed that spearcons require almost no learning to comprehend, and they are as easy to learn as speech. Due to spearcons' effectiveness, they have been widely used for various warnings in manually driven vehicles (e.g., Jeon et al., 2022; Šabić et al., 2021; Sabic et al., 2017). However, the potential of using spearcons as TORs in automated vehicles is relatively underexplored. Only Jeon (2019) explored the effectiveness of spearcon as a TOR in comparison with the earcon and speech, and the author suggested that the spearcon is not a suitable TOR for getting drivers back into the control loop due to its longer takeover time compared to that of the earcon.

1.2.2. Non-speech-based warnings

Another category of auditory warnings is non-speech-based warnings, which mainly include earcons and auditory icons (Absar and Guastavino, 2008). An earcon is an abstract synthetic sound that has an arbitrary relationship with the object or the action it represents (Blattner et al., 1989), such as beeps. It is easy to manipulate and change an earcon by modulating its acoustic parameters, such as its pitch, timbre, and rhythm. Earcons' effectiveness as in-vehicle warnings has been demonstrated in previous studies (Landry et al., 2015, 2019). Many studies have also used earcons as TORs in automated vehicles (e.g., Bazilinskyy et al., 2018; Beattie et al., 2015; Petermeijer et al., 2017b). However, due to the arbitrary mapping relationship of earcons, it is relatively difficult for users to remember their meaning and easy to be confused (Dingler et al., 2008). Hence, earcons may lead to a memory and training burden for drivers.

Auditory icons refer to natural sounds with nomic or metaphorical mapping to the information being represented (Gaver, 1989). For example, the sound of crumpling paper occurs when a file is dragged into the recycle bin in the Mac operating system, indicating that the file has been deleted. Auditory icons take advantage of users' prior knowledge to create a natural and direct mapping so that users need no or little training to learn and remember their meanings (Belz et al., 1999; Dingler et al., 2008). Moreover, auditory icons can induce faster responses

in users (Cabral and Remijn, 2019). This characteristic is important in in-vehicle interface design, where a late response may result in a serious accident. Therefore, in the manual driving field, several studies have proposed the use of auditory icons as in-vehicle warnings (Bakowski et al., 2015; Larsson et al., 2009; McKeown and Isherwood, 2007; Song et al., 2022). Nevertheless, few studies have investigated the effectiveness of using auditory icons as TORs in automated vehicles.

1.2.3. Designing effective takeover requests using an appropriate type of auditory warning

With the approaching of the automated driving era, it is important to design an effective auditory takeover system, and choosing an appropriate auditory warning for the TOR is necessary. In automated vehicles, drivers must resume control within a reasonable amount of time after a TOR has been issued to ensure their driving safety (Zeeb et al., 2015). Speech and earcon have often been used as TORs in previous studies (e.g., Forster et al., 2017; Ma et al., 2021; Wang et al., 2022; Bazilinskyy et al., 2018; Beattie et al., 2015; Petermeijer et al., 2017b), whereas auditory icons and spearcons are relatively underused as TORs. Nevertheless, according to previous studies on manual driving, auditory icons and spearcons as warnings can lead to a shorter response time compared with earcons and speech, respectively. For example, Sabic et al. (2017) assessed drivers' responses to speech and spearcons as in-vehicle auditory warnings in the background of manual driving. They found that spearcons lead to a faster response time than speech. McKeown and Isherwood (2007) and Song et al. (2022) also compared auditory icons with earcons in the vehicle warning system, and their results showed that auditory icons result in a faster response time and better driving performance than earcons. Hence, auditory icons and spearcons may have great application potential as TORs.

However, generalizing the results above for manual driving to the field of automated driving may be problematic. Unlike a manually driven vehicle, drivers in an automated vehicle are free from needing to drive, so they can engage in NDRTs and can remain outside of the control loop. That is, during automated driving, the NDRT is the primary task, and monitoring the state of the automated system is a secondary task. When drivers are required to take over control of the vehicle, they must cease the NDRT and take over driving as their primary task. This is very different from manual driving, wherein the primary task is always driving. It has been found that drivers in automated vehicles have low situational awareness, and their workload is higher when they are required to take over compared to during manual driving (see de Winter et al., 2014 for a review). Furthermore, Van der Heiden et al. (2021) revealed that engaging in NDRTs during automated driving decrease auditory susceptibility. Due to the differences between manual and automated driving, the results for manual driving may not be generalizable to automated driving. Hence, it is necessary to conduct research within the background of automated driving to explore the potential and effectiveness of various auditory warnings as TORs. Nevertheless, few studies have been conducted on this issue.

1.3. The impact of auditory non-driving-related tasks on the effectiveness of takeover requests

Since drivers are extremely willing to engage in various NDRTs during automated driving (Pfleger et al., 2016), such as listening to music, texting, talking, etc., drivers' visual or auditory modality is often occupied by various NDRTs. NDRTs' modality may affect auditory TORs' effectiveness. In accordance with multiple resources theory (Wickens, 2002), drivers' cognitive resources in each perceptual modality are limited and independent. The information in separate modalities can be simultaneously processed and will not result in resource competition. That is, if drivers are engaged in visual NDRTs, the NDRTs do not compete for the auditory cognitive resources needed for the perception and processing of auditory TORs. Hence, visual NDRTs may have little influence on auditory TORs' effectiveness.

However, if drivers are occupied with auditory NDRTs, the NDRT will compete with auditory TORs for auditory cognitive resources, which, in turn, would influence the auditory TOR's effectiveness. In manual driving research, several studies have indicated that it is necessary to consider the detrimental impact of other in-vehicle auditory tasks on the reception of auditory warnings when designing a warning system (Merchel and Altinsoy, 2020; Nees and Walker, 2011; Šabić et al., 2021). A concurrent auditory task can interfere with and even mask the target auditory warning (Nees and Walker, 2011). Moreover, various types of auditory warnings have different susceptibilities to the interference of concurrent auditory tasks (Glatz et al., 2018). For example, Vilimek and Hempel (2005) and Bonebright and Nees (2009) found that speech interferes with concurrent auditory memory tasks, whereas auditory icons and earcons do not.

It can be inferred that when drivers are performing auditory NDRTs, fewer auditory cognitive resources are available for processing auditory TORs than when they are performing visual NDRTs due to the resource competition within the same modality. Speech-based warnings as TORs may not be very suitable because drivers need more auditory cognitive resources to understand their meanings than non-speech-based warnings (Simpson and Marchionda-Frost, 1984; Wu and Boyle, 2021). Therefore, it would be appropriate to use more instinctive warning types as TORs, such as auditory icons, when drivers are engaged in auditory NDRTs. This can free up some of the auditory cognitive load and prompt drivers to respond more quickly. However, this would not apply when drivers are engaged in visual NDRTs. Due to less resource competition, drivers would have sufficient auditory resources to process any type of auditory warning. In these circumstances, fewer differences may exist among the various auditory warnings. Hence, it is important to consider the effect of the NDRT modality when designing auditory TORs. Petermeijer et al. 2017b and Yoon et al. (2019) investigated the moderating effect of the NDRT modality on the impact of the TOR modality on takeover performance. However, these two studies mainly focused on the TOR modality and only used one type of auditory TOR. Therefore, the present study aimed to concentrate on the auditory TOR and further investigate the impact of the NDRT modality (visual or auditory) on various types of auditory TORs.

1.4. Research gaps and the aims of the present study

Designing an effective auditory TOR is critical to ensure automated driving safety. Previous related studies have mostly used speech or earcons as TORs. Auditory icons and spearcons are relatively underused, although they have great potential as TOR candidates. Moreover, which type of auditory warning is most suitable for usage as a TOR remains inconclusive. Online surveys have been conducted to investigate people's preferences for various types of auditory TORs (Bazilinskyy et al., 2018; Bazilinskyy and Winter, 2015). However, these subjective ratings are not equivalent to objective driving performance. In addition, Jeon (2019) conducted an experiment to compare earcons, speech, and spearcons as TORs. Nevertheless, the study did not consider the potential of auditory icons, nor did it collect data for takeover quality, which is also of great importance when assessing takeover performance apart from takeover timing metrics (Cao et al., 2021; Zeeb et al., 2016). Therefore, we conducted a simulated driving experiment to systematically explore the effects of four types of auditory TORs using comprehensive metrics, including takeover timing and quality performance, as well as subjective preferences. Furthermore, few previous studies have considered the potential interference of auditory NDRTs with different types of auditory TORs. Hence, the present study also considered the modality of NDRTs (visual or auditory) to compare the effects of various auditory TORs. To summarize, the present study aimed to answer the following research questions:

- 1a) Are auditory icons and spearcons more beneficial for takeover performance than earcons and speech, respectively, when used as TORs in automated vehicles?
- 1b) Which of the four types of warnings is the most effective as a TOR?
- 2a) Do auditory NDRTs have a detrimental impact on the effectiveness of various auditory warnings, compared with visual NDRTs?
- 2b) Which type of warning is the most appropriate for use as a TOR under the visual and auditory NDRT conditions?

2. Methods

2.1. Participants

Thirty-two participants from Zhejiang University (26 male, 6 female) were recruited for this experiment. All of them held a valid Chinese driver's license at the time of this study. Their ages ranged from 20 to 27 years, with an average age of 22.78 years ($SE = 0.36$). Their average driving experience was 2.75 years ($SE = 0.36$), and their average total mileage was 5400.00 km ($SE = 3208.73$). All of the participants had normal vision and hearing and did not suffer from any driving-related diseases. Participants were randomly assigned to two groups corresponding to two NDRT conditions (see Section 2.6 for more details). No significant difference was observed in terms of age ($t_{(30)} = -.95, p = .35$), years of driving ($t_{(30)} = -1.43, p = .16$), and total mileage ($t_{(30)} = -1.34, p = .19$) between the two groups.

2.2. Apparatus

The driving simulator consisted of driving software (STISIMDRIVE M1000-R), an operating system (Logitech G29) including brake pedals, a steering wheel, and an accelerator, an adjustable seat, two desktops (Nvidia GeForce RTX 3080 [10 GB], Intel Core i7-10700K), a 55-inch display with 3840×2160 pixels resolution, and a pair of speakers. The distance between the screen and the participants was approximately 1.1 m. The driving simulator software was run on computer A to create the driving scenarios, and the scenarios were presented on the 55-inch screen. We used another computer (computer B) to run a program written in Python to collect and calculate the driving data from computer A in real time. The data communication between these two computers was realized via serial communication. The experimental setup is shown in Fig. 1.

2.3. Auditory takeover displays

This study aimed to explore the effects of different types of auditory TORs on takeover performance. The TOR types in the experiments included speech, spearcon, auditory icon, and earcon. The engine noise



Fig. 1. The experimental setup.

during the experiment was approximately 50 dB. To prevent the masking effect of engine noise on takeover warnings, all auditory warnings were set at 70 dB (Patterson, 1982).

(1) Speech and earcons

The speech warning translated from Chinese was “Danger ahead; please take over.” which was presented in a female voice. It was generated with online text-to-speech software (<https://www.text-to-speech.cn/>), and its length was approximately 1900 ms. The earcon used in the present study followed the design guidelines of Brewster et al. (1993). It was a single pair of 240 ms beeps at 2700 Hz, with a 100 ms interval between the two beeps, the effectiveness as a TOR of which was demonstrated in the study of Petermeijer et al. (2017b). The earcon was created using Audacity software.

(2) Spearcons

The spearcon was first introduced by Walker et al. (2006) as an alternative to speech warnings. It is a compressed version of speech warnings: It has an increased tempo but unchanged pitch. Due to the shortened nature of spearcons, users can comprehend their meaning in a shorter time and complete tasks more quickly and efficiently. There are usually two compression methods, namely, linear compression and compression to the duration of certain words. The latter leads to more diverse compression degrees depending on the length of the original speech. If the original speech is longer, the compression degree will be larger. However, this method may make it more difficult to rigorously control the compression degree and so has been accepted infrequently in previous studies (Walker et al., 2013; Sabic et al., 2017; Šabić et al., 2021). Hence, we chose to compress speech linearly in the present study. Moreover, it is important to ensure that the compression degree is not too large for users to comprehend the meaning of the warnings, although a large compression degree would theoretically reduce the time that users would require to hear the complete content. The compression degree should be set to an appropriate level. Sabic and Chen (2016) investigated various linear compression degrees (ranging from 20% to 100% at an interval of 10%) of spearcons. They found that participants struggled to identify spearcons when they were compressed to less than 40% of the original speech duration, but 40% is the ideal compression degree to guarantee accuracy and efficiency. Therefore, the length of the spearcon was set at 40% of the speech duration in the present study, which was approximately 760 ms.

(3) Auditory icons

With respect to auditory icons, Mynatt (1994) and Cabral and Remijn (2019) proposed four key points that should be considered when designing useful and effective auditory icons: a) The identification of auditory icons: When an auditory icon is not easy for users to identify, its usefulness cannot be guaranteed. Users’ familiarity with the sound of auditory icons is important for the identification of auditory icons. In addition, the physical characteristics of auditory icons play an important role in their identification. b) The conceptual mapping of auditory icons: To ensure the usability of auditory icons, their conceptual mapping is of great significance. As described in Section 1.2.2, the sound should have a natural and instinctive relationship with the referent to improve mapping, and this principle is at the core of designing auditory icons. c) The physical parameters of auditory icons: Designers should also consider the physical parameters of auditory icons to facilitate their usability, such as sound duration and intensity. These characteristics directly impact the identification of auditory icons. d) User preferences for auditory icons: User preferences also influence the usability of auditory icons. Previous studies have revealed users prefer icons with a strong relationship to their referents (Wang et al., 2012; Höferlin et al., 2011). Moreover, users’ emotional responses to icons would affect their

preferences and should be considered when designing icons.

Taking these factors into consideration, the sound of a traffic collision was used as the auditory icon to inform drivers to take over in the present study. First, the sound is extremely common, so people are usually familiar with it, making it easily identifiable for drivers. Second, the sound is naturally and instinctively mapped to a takeover event since the host vehicle would collide with potential hazards ahead if the driver does not intervene in the takeover situation. Therefore, the sound meets a necessary design point of conceptual mapping. Third, with respect to the physical parameters of the auditory icon, the sound was set at approximately 70 dB, and its duration was approximately 2100 ms. In accordance with Patterson’s (1982) guidelines, the intensity of the warning was set to 20 dB higher than the background engine noise (approximately 50 dB) to guarantee that the warning was not masked and drivers could easily identify it. Notably, the intensity of the other three types of TORs was set at 70 dB to prevent the confounding effect of intensity. Fourth, users’ preferences for the sound of a traffic collision as a warning of potential hazards ahead have been examined in a previous study. Specifically, Song et al. (2022) investigated the effects of the sounds of a traffic collision and glass breaking to warn drivers of hazards on the road, and the results showed that the traffic collision sound had greater efficacy and annoyance ratings and thus was preferred by drivers.

In summary, the sound of a traffic collision meets the four key design points that Mynatt (1994) and Cabral and Remijn (2019) proposed for auditory icons that warn drivers to take over. Its effectiveness has also been validated in several studies (Cabral and Remijn, 2019; McKeown and Isherwood, 2007; Song et al., 2022). Therefore, we used it as the auditory icon in the present study.

2.4. Non-driving-related tasks

The present study used the N-back task as the NDRT, which has been widely used in previous automated driving studies (e.g., Jeong and Liu, 2019; Petermeijer et al., 2017b; Radlmayr et al., 2014). Participants in the experiment performed either visual or auditory 2-back tasks, as Mehler et al. (2011) specified.

In the 2-back tasks, a series of random numbers appeared at pre-defined intervals (2 s) on an iPad, which had a 10.2-inch screen. Participants could see only one number once. The visual and auditory numbers were both presented for about 500 ms, followed by a 1500 ms blank display, during which there was nothing to see or hear. Hence, participants were exposed to each number for 500 ms to memorize it. When the current number matched the 2-back number, participants had to respond by tapping on the screen of the iPad, as shown in Fig. 2. The iPad was fixed at approximately 35 cm to the right of the steering wheel and 30° below participants’ line of driving sight. As long as the automated driving system was activated, participants were required to engage in the N-back task until TORs were issued. The automated driving lasted 120–150 s. Hence, participants were required to complete 60 to 75 N-back trials in total, and the target number that participants needed to hit took around 20% of those trials. Under the visual NDRT condition, the numbers in the 2-back task were displayed on the screen, thereby occupying participants’ visual modality to perform the task. Under the auditory NDRT condition, a female voice presented the

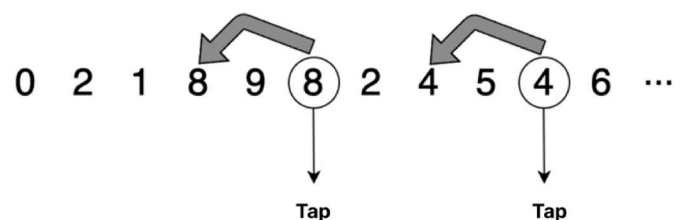


Fig. 2. An illustration of the 2-back tasks.

numbers. Participants depended on their auditory modality to perform the task. All participants were randomly assigned to either the visual or the auditory NDRT group.

Before the formal experiment, we recruited 10 participants to conduct a preliminary experiment to test the workload of the visual and auditory N-back tasks, and a seven-point Likert scale was used to assess mental workload. The sequence of the two N-back tasks was balanced across participants. The paired *t*-test showed that there was no significant difference in the workload of the visual and auditory tasks ($t_{(9)} = .21$, $p = .84$, $\text{Mean}_{(\text{visual})} = 5.30 \pm 1.06$, $\text{Mean}_{(\text{auditory})} = 5.40 \pm 1.17$).

2.5. Driving scenarios

All driving scenarios were built using STISIM software. In a sunny environment, the automated vehicle was driven in the middle lane of a three-lane highway at a speed of 100 km/h. Each lane was 5 m wide.

The experiment comprised four types of takeover events, namely a traffic accident, obstacles, a work zone, and a broken-down truck. These takeover events were adopted from the study of Zhang et al. (2021). Brief descriptions of these events are provided in Table 1. The hazards for each type of event could occupy either the left and front lanes or the right and front lanes, and they were completely symmetrical relative to the central axis of the three lanes. Hence, by combining the specific location of hazards for each takeover event type, there were a total of eight takeover scenarios.

After the automated driving system was activated, drivers were engaged in NDRTs during automated driving. When the vehicle was 222 m away from the hazard (approximately 8 s), the takeover request was issued (Damböck et al., 2012). Drivers had to take over control of the automated vehicle. After issuing the TORs, the drivers had to steer their vehicles to the left or right to avoid a collision. During the takeover process, there were no other vehicles on the road since these could have an impact on their takeover responses.

Table 1
Descriptions of the four types of takeover events.

Scenario	Screenshot	Description
Traffic accident		This scenario included three collided cars on the main road.
Obstacles		In this scenario, sandpiles and some traffic cones were situated on the main road. There was also a warning sign in front of the obstacles.
Work zone		A work zone occupied the main road, comprising two sandpiles and a bulldozer.
Broken-down truck		In this scenario, a truck was broken down on the road. Several boxes had fallen off the truck and were blocking two of the lanes.

2.6. Experimental design

The experiment adopted a 2 (NDRT modality) $\times$ 4 (TOR type) mixed design. The NDRT modality (visual vs. auditory) was manipulated as a between-subject variable. Participants were randomly assigned to the visual or auditory NDRT group. In the visual NDRT group, participants were required to perform visual N-back tasks during automated driving, while participants in the auditory group were required to perform auditory N-back tasks. The TOR type was treated as a within-subject variable and comprised four types of TORs, namely auditory icon, earcon, speech, and spearcon.

Participants in each group completed four blocks corresponding to the four TOR types, and each block comprised two trials of the same TOR type. The sequence of four blocks was balanced using a Latin square. The takeover scenario for each trial was randomly chosen from the scenarios described above.

2.7. Measurement

2.7.1. Objective metrics

Previous studies have shown that to obtain a complete overview of takeover performance, objective metrics should incorporate both timing and quality aspects (Cao et al., 2021; Zeeb et al., 2016). Table 2 shows the specific metrics used in the present study. The objective metrics were recorded and calculated from the issuing of TORs to the moment at which the ego vehicle passed the location of the hazard.

2.7.1.1. Takeover timing. In accordance with the study of Cao et al. (2021), two metrics related to takeover timing were recorded in this experiment:

- (1) Takeover time: defined as the time interval between the issuing of the TOR and the driver's first control input (moving the steering wheel more than 2° or depressing the brake pedal further than 10% (Gold et al., 2013)).
- (2) Lane change time: defined as the time interval between the issuing of the TOR and the ego vehicle's deviation of greater than 2.5 m (half the width of the lane) from the road's center (Petermeijer et al., 2017b).

2.7.1.2. Takeover quality. Two metrics were recorded to assess takeover quality in this experiment:

Table 2
Summary of the dependent variables recorded in the experiment.

Type	Specific metric	Unit	Description
Takeover timing	Takeover time	s	The time interval between the TOR occurrence and the driver's first maneuver input (moving the steering wheel more than 2° or depressing the brake pedal further than 10 %).
	Lane change time	s	The time interval between the TOR occurrence and the ego vehicle's deviation of greater than half the lane width.
Takeover quality	Maximum resultant acceleration	m/s ²	The maximum value of resultant acceleration during the takeover process.
	Minimum TTC	s	The TTC with the upcoming hazard when the ego vehicle deviated from the road center by more than half the lane width.
Subjective metrics	Workload	None	The workload of the takeover (measured using the NASA-TLX scale)
	Preference rank	None	The preference rank for the four TOR types

- (1) Maximum resultant acceleration: defined as the maximum value of acceleration during the takeover process. A smaller acceleration indicates smoother control transfer and greater takeover quality.
- (2) Minimum time to collision (Min-TTC): defined as the TTC with the upcoming hazard when the ego vehicle deviated from the road's center by more than half of the lane width (Gold et al., 2016). A greater value for this metric indicates a safer takeover maneuver.

2.7.2. Subjective metrics

The NASA Task Load Index (NASA-TLX) scale was used to assess drivers' workload during the takeover when using various TORs. The NASA-TLX includes six dimensions, namely mental demand, physical demand, temporal demand, performance, effort, and frustration. Each dimension is rated from 0 to 100 at an interval of 5 points. The overall workload is calculated by using the mean score of the six dimensions (Hart and Staveland, 1988).

At the end of the experiment, participants were also required to give their preference rank for the four auditory TOR types that they experienced. The subjective questionnaires were digital (created using the Wenjuanxing platform [https://www.wjx.cn/]). Participants were required to complete the questionnaires on the iPad provided.

2.8. Procedure

After arrival, participants were welcomed. They signed the informed consent and completed a demographic questionnaire about their gender, age, and driving experience. Then, they were provided with an introduction to the purpose of the experiment and the different types of auditory TORs. The participants were randomly assigned to either the visual or auditory NDRT group. The whole experiment included a practice session and a formal experiment session.

During the practical session, participants were first given the instructions for either the visual or auditory N-back task. They then had the opportunity to practice the N-back tasks for approximately 2 min. Then, they adjusted their seat to a comfortable position and drove the vehicle in manual mode for approximately 5 min to familiarize themselves with the simulated driving system. Afterward, participants were required to complete an automated drive to familiarize themselves with the automated driving system and takeover process. During the automated driving, they were required to perform either the visual or auditory N-back task. The takeover TOR for the practice drive was randomly chosen from the four types of auditory TORs.

During the formal experiment session, participants completed four blocks corresponding to the four types of auditory warnings (speech, spearcon, earcon, auditory icon). The sequence of the four blocks was balanced using a Latin square. Each block included two trials. Hence, each participant was required to complete a total of eight trials for the formal experiment. In each trial, participants performed the NDRT task for 2–2.5 min in order to prevent them from anticipating the takeover event. Between each block, there was a short break. Participants completed the NASA-TLX rating to evaluate their workload during the takeover process for the previous two trials. When completing the four blocks, they were required to rank the four TOR types according to their preference. The whole experiment lasted approximately 50 min, and after the experiment, participants were compensated with 40 yuan for their participation.

3. Results

IBM SPSS 25.0 was used for data analysis. Participants in the present study were required to complete two trials corresponding to each experimental condition and rate their workload for the takeover once after completing the two trials. Hence, the data construct for workload was different from that of objective performance. Therefore, we

conducted a 2×4 mixed-design MANOVA test within the framework of a general linear model for takeover performance (i.e., takeover time, lane change time, maximum resultant acceleration, and min-TTC) and workload (i.e., overall workload and its score in six dimensions), respectively. The NDRT type was treated as a between-subject factor and the auditory TOR type as a within-subject factor. In terms of the preference rank data, we used the Friedman test, which is a non-parametric statistical method for ordinal data with repeated measures (Friedman, 1937). The Wilcoxon signed-rank test was employed for the post hoc test. Since the Friedman test has limitations when testing interaction effects, we examined the effect of the auditory TOR type on the preference rank for the visual and auditory NDRT conditions. All significance levels were set at 0.05. The standard errors of the within-subject variables were calculated using the method proposed by Cousineau and O'Brien (2014).

3.1. Takeover performance

The results of the MANOVA for the combined dependent variables showed a significant effect of NDRT ($F_{(4, 218)} = 5.70, p < .001$, Wilks's $\lambda = .91, \eta_p^2 = .10$) and TOR type ($F_{(12, 577)} = 2.24, p = .009$, Wilks's $\lambda = .89, \eta_p^2 = .04$). Their interaction effect was insignificant ($F_{(12, 577)} = .77, p = .68$, Wilks's $\lambda = .96, \eta_p^2 = .01$).

3.1.1. Takeover time

Table 3 shows the results of the interaction and main effects of the NDRT modality and TOR type on takeover time. A statistically significant difference between the visual and auditory NDRTs was observed. The auditory NDRT (Mean = 3.02 ± 0.09 s) resulted in a longer takeover time than the visual NDRT (Mean = 2.76 ± 0.09 s, $p = .01$). In addition, the main effect of the TOR type was significant, and the post hoc test results showed that the takeover time for the earcon TOR (Mean = 2.53 ± 0.12 s) was significantly shorter than for the spearcon (Mean = 3.09 ± 0.15 s, $p = .001$) and speech (Mean = 3.12 ± 0.13 s, $p = .002$) TORs. The auditory icon TORs (Mean = 2.83 ± 0.13 s) showed no significant difference from the other three types of TORs. Although the interaction effect of the NDRT modality and TOR type was insignificant, there was also an obvious tendency. When drivers engaged in the auditory NDRT, there was no significant difference among the four types of TORs. However, when engaging in the visual NDRT, the earcon TOR had a significantly shorter takeover time than the speech ($p = .004$) and spearcon ($p = .001$) TORs. Fig. 3 shows the detailed results of the various TORs under the visual and auditory conditions.

Table 3

Results for the interaction and main effects of the non-driving-related task modality (NDRT) and takeover request type (Type) on takeover performance.

Effect	<i>F</i>	<i>df</i>	<i>p</i>	η_p^2
Takeover time				
NDRT	6.49	1, 221	.01	.03
Type	4.92	3, 221	.003	.06
NDRT*Type	1.04	3, 221	.37	.01
Lane change time				
NDRT	4.49	1, 221	.04	.02
Type	3.20	3, 221	.02	.04
NDRT*Type	.31	3, 221	.82	.004
Maximum resultant acceleration				
NDRT	8.15	1, 221	.005	.04
Type	.35	3, 221	.79	.01
NDRT*Type	.01	3, 221	1.00	.001
Min-TTC				
NDRT	13.34	1, 221	< .001	.06
Type	5.11	3, 221	.002	.07
NDRT*Type	.16	3, 221	.92	.002

Notes. Significant *p* values are highlighted in bold.

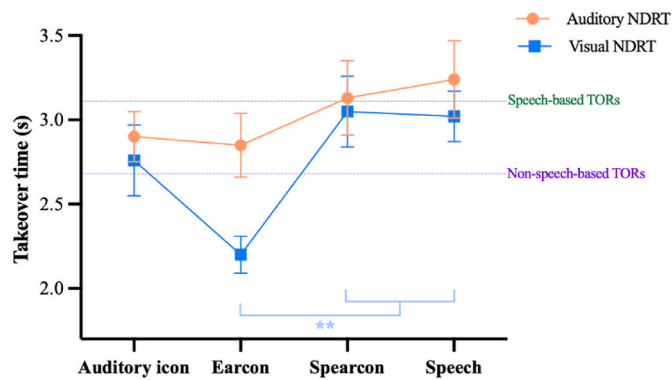


Fig. 3. Means of takeover time per takeover request type under the auditory and visual non-driving-related task conditions. The error bars represent the standard errors of the means. The green dotted line indicates the mean value of speech-based takeover requests, including speech and spearcons, and the purple dotted line indicates the mean value of non-speech-based takeover requests, including earcons and auditory icons. (Notes. $**p < .01$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.1.2. Lane change time

Table 3 shows the results of the interaction and main effects of the NDRT modality and TOR type on lane change time. The main effects of the NDRT modality and TOR type were significant. The post hoc test showed that the auditory NDRT (Mean = 5.12 ± 0.10 s) resulted in a longer lane change time than the visual NDRT (Mean = 4.82 ± 0.08 s, $p = .04$). The lane change time for the speech TOR (Mean = 5.26 ± 0.14 s) was also longer than for the earcon (Mean = 4.80 ± 0.15 s, $p = .01$) and auditory icon (Mean = 4.81 ± 0.14 s, $p = .01$) TORs. Although the interaction effect of the TOR type and NDRT modality did not reach statistical significance, there was still a distinct pattern. That is, under the visual NDRT condition, the speech TOR resulted in a longer lane change time than the auditory icon ($p = .05$) and earcon ($p = .02$) TORs, whereas there was no significant difference among the various TORs under the auditory NDRT condition (see Fig. 4).

3.1.3. Maximum resultant acceleration

Only the main effect of the NDRT modality was significant. The auditory NDRT (Mean = 2.24 ± 0.10 m/s²) was associated with smaller maximum resultant acceleration than the visual NDRT (Mean = $2.75 \pm$

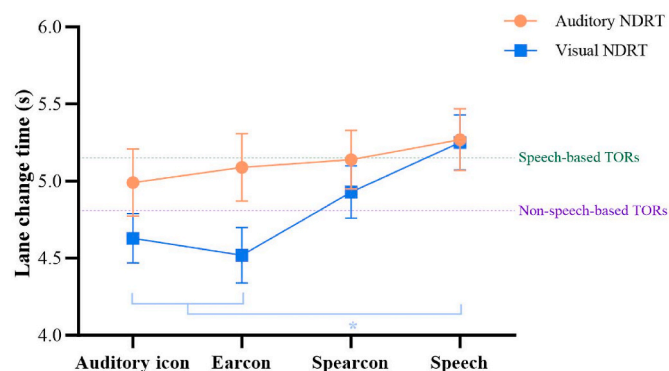


Fig. 4. Means of lane change time per takeover request type under the auditory and visual non-driving-related task conditions. The error bars represent the standard errors of the means. The green dotted line indicates the mean value of speech-based takeover requests, including speech and spearcons, and the purple dotted line indicates the mean value of non-speech-based takeover requests, including earcons and auditory icons. (Notes. $*p < .05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

0.13 m/s², $p = .005$). Neither the main effect of the TOR type nor its interaction effect with the NDRT modality on the maximum resultant acceleration was significant, as shown in Table 3. More detailed results are shown in Fig. 5.

3.1.4. Minimum time to collision

The NDRT modality and TOR type had a significant effect on the min-TTC, but their interaction effect was insignificant, as shown in Table 3. The post hoc test for the NDRT modality also showed that the min-TTC under the auditory NDRT condition (Mean = 3.49 ± 0.08 s) was shorter than under the visual NDRT condition (Mean = 3.90 ± 0.08 s, $p < .001$). Further, the post hoc test for the TOR type showed that the min-TTC was significantly shorter with the speech TOR (Mean = 3.38 ± 0.13 s) than with the auditory icon (Mean = 3.75 ± 0.13 s, $p = .02$) and earcon (Mean = 4.00 ± 0.11 s, $p < .001$) TORs. The min-TTC with the spearcon TOR (Mean = 3.65 ± 0.12 s) was also shorter than with the earcon TOR ($p = .02$). Fig. 6 shows the detailed results of the various TORs under the visual and auditory NDRT conditions.

3.2. Subjective metrics

3.2.1. Workload

According to the results of the MANOVA, neither a main effect of the NDRT modality ($F_{(6, 133)} = .97$, $p = .45$, Wilks's $\lambda = .95$, $\eta_p^2 = .05$) and TOR type ($F_{(18, 320)} = 1.18$, $p = .28$, Wilks's $\lambda = .83$, $\eta_p^2 = .06$) nor an interaction effect ($F_{(18, 320)} = .62$, $p = .88$, Wilks's $\lambda = .91$, $\eta_p^2 = .03$) on the combined dependent variables was observed.

In addition, neither a main effect of the NDRT modality and TOR type nor an interaction effect on the overall score or dimensional scores was found (see Table 4). Fig. 7 shows the detailed results of the overall score, and Table 5 shows the detailed results of the six dimensions.

3.2.2. Preference ranks for the takeover requests

Table 6 shows the preference rank medians for the various TOR types under the visual and auditory NDRT conditions. Under the visual NDRT condition, the Friedman test showed a significant effect of TOR type on the preference rank ($\chi^2(3) = 21.22$, $p < .001$). The post hoc test revealed that the speech TOR (Median = 1) resulted in a greater preference rank than the spearcon (Median = 3) and auditory icon (Median = 4) TORs. No other significant pairwise comparisons were found.

Under the auditory NDRT condition, the Friedman test also showed a significant main effect of TOR type ($\chi^2(3) = 13.13$, $p = .004$). Additionally, the post hoc test revealed a significant difference between the

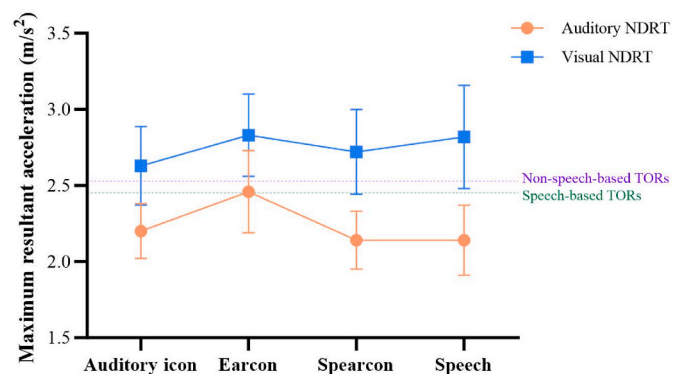


Fig. 5. Means of maximum resultant acceleration per takeover request type under the auditory and visual non-driving-related task conditions. The error bars represent the standard errors of the means. The green dotted line indicates the mean value of speech-based takeover requests, including speech and spearcons, and the purple dotted line indicates the mean value of non-speech-based takeover requests, including earcons and auditory icons. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

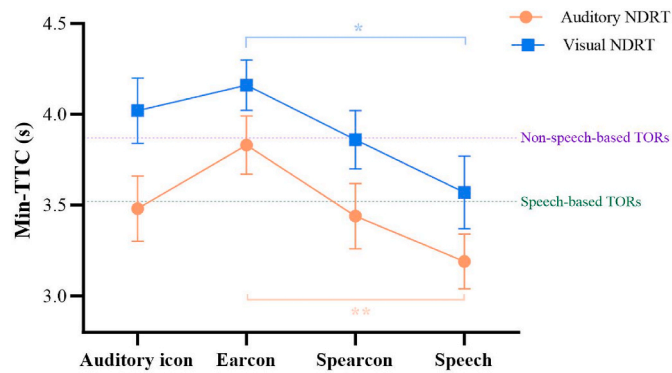


Fig. 6. Means of minimum time to collision per takeover request type under the auditory and visual non-driving-related task conditions. The error bars represent the standard errors of the means. The green dotted line indicates the mean value of speech-based takeover requests, including speech and spearcons, and the purple dotted line indicates the mean value of non-speech-based takeover requests, including earcons and auditory icons. (Notes. * $p < .05$, ** $p < .01$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 4

Results for the interaction and main effects of the non-driving-related task modality (NDRT) and takeover request type (Type) on takeover workload.

Effect	<i>F</i>	<i>df</i>	<i>p</i>	η_p^2
Overall				
NDRT	3.72	1, 118	.06	.03
Type	2.07	3, 118	.11	.05
NDRT*Type	.34	3, 118	.80	.01
Mental				
NDRT	3.17	1, 118	.08	.03
Type	1.27	3, 118	.29	.03
NDRT*Type	.56	3, 118	.64	.01
Physical				
NDRT	1.59	1, 118	.21	.01
Type	.88	3, 118	.45	.02
NDRT*Type	.06	3, 118	.98	.001
Temporal				
NDRT	3.11	1, 118	.08	.03
Type	2.22	3, 118	.09	.05
NDRT*Type	.25	3, 118	.86	.01
Effort				
NDRT	1.06	1, 118	.31	.01
Type	.63	3, 118	.60	.02
NDRT*Type	.10	3, 118	.96	.003
Performance				
NDRT	1.15	1, 118	.29	.01
Type	2.60	3, 118	.06	.06
NDRT*Type	1.05	3, 118	.37	.03
Frustration				
NDRT	3.89	1, 118	.05	.03
Type	1.96	3, 118	.12	.05
NDRT*Type	.62	3, 118	.60	.02

speech (Median = 1.5) and auditory icon (Median = 3.5) TORs. No other significant pairwise comparisons were found.

4. Discussion

Speech and earcons have been widely used as auditory TORs to prompt drivers to return to the control loop in previous studies, but the potential of using auditory icons and spearcons as TORs has been relatively underexplored. Hence, the present study explored the effects of various auditory TORs on takeover performance and subjective preference. Moreover, we also considered the impact of the NDRT modality on various auditory TORs due to its potential interference effect. The present study's findings have some practical implications for the design of

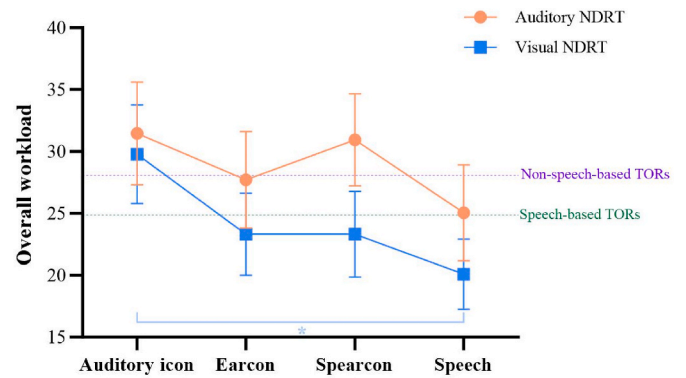


Fig. 7. Means of overall workload during takeover per takeover request type under the auditory and visual non-driving-related task conditions. The error bars represent the standard errors of the means. The green dotted line indicates the mean value of speech-based takeover requests, including speech and spearcons, and the purple dotted line indicates the mean value of non-speech-based takeover requests, including earcons and auditory icons. (Notes. * $p < .05$). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

auditory takeover systems in automated vehicles.

4.1. The effects of various auditory takeover requests

Although the four types of warnings in the present study have been widely investigated and used in the manual driving field, designers should be cautious when generalizing them to the automated driving field as TORs due to the difference between the two fields. That is, with manual driving, drivers must continuously focus on the road ahead while driving. Warnings of potential hazards tend to be more easily detectable and understandable for manual drivers due to their consistent “in-the-control-loop” state (de Winter et al., 2014). However, drivers are allowed to engage in NDRTs in automated vehicles, so they are out of the control loop during automated driving. When TORs occur, drivers' perception and cognitive processing of TORs are usually impacted by NDRTs because their cognitive resources are devoted to NDRTs (Morales-Alvarez et al., 2020; de Winter et al., 2014). These differences should be considered when designing TORs for automated vehicles.

Bazilinsky and Winter (2015, 2018) investigated people's preferences for various auditory TORs via crowdsourcing surveys and found that people preferred speech more than abstract sounds. However, a greater subjective preference does not indicate better takeover performance. Therefore, the present study conducted a simulated driving experiment to compare various auditory TORs while considering the potential of auditory icons and spearcons as TORs. The results showed that non-speech-based TORs (earcon and auditory icon) can lead to shorter takeover times and lane change times than speech-based TORs (speech and spearcon). This is probably because speech-based TORs have a relatively slower presentation rate than non-speech-based TORs. That is, drivers need to listen to all of the speech content before taking over control of the vehicle (Jeon et al., 2022), resulting in a delayed response.

For the non-speech-based TORs, it is interesting to find that drivers responded to earcon TORs more quickly than to auditory icon TORs, though this difference did not reach statistical significance. Previous studies in the manual driving field have found that using auditory icons as collision warnings can lead to faster response times than earcons (Belz et al., 1999; McKeown and Isherwood, 2007; Song et al., 2022). They claimed that this is because auditory icons reduce cognitive processing times due to their direct nomic mapping to the information being represented. However, in the present study, auditory icons as TORs resulted in slower takeover times than for earcons. Given that the acoustic parameters of the earcons and the type of auditory icons used in the present

Table 5

The mean scores and standard errors of each dimension of the NASA Task Load Index as a function of the non-driving-related task modality and takeover request type.

	Auditory NDRT				Visual NDRT			
	Auditory icon	Earcon	Spearcon	Speech	Auditory icon	Earcon	Spearcon	Speech
Overall	31.46 ± 4.16	27.71 ± 3.91	30.94 ± 3.72	25.05 ± 3.87	29.79 ± 3.98	23.33 ± 3.33	23.33 ± 3.46	20.01 ± 2.84
Mental	36.88 ± 5.21	29.38 ± 4.88	38.75 ± 5.12	30.94 ± 5.85	34.06 ± 5.48	28.75 ± 5.07	26.88 ± 5.02	25.00 ± 4.50
Physical	27.19 ± 4.94	25.94 ± 4.84	25.94 ± 4.38	22.19 ± 3.91	25.94 ± 5.22	21.88 ± 4.62	20.94 ± 3.99	18.44 ± 4.56
Temporal	43.75 ± 7.91	42.19 ± 6.62	39.38 ± 5.70	28.75 ± 5.36	38.13 ± 6.32	31.56 ± 6.32	30.94 ± 6.82	23.44 ± 4.04
Effort	29.38 ± 5.75	27.19 ± 5.96	28.75 ± 4.79	26.88 ± 5.70	30.00 ± 5.40	24.38 ± 4.28	25.00 ± 4.01	21.88 ± 3.37
Performance	23.13 ± 3.90	17.50 ± 2.36	27.81 ± 4.15	19.33 ± 3.33	25.00 ± 4.74	17.50 ± 2.04	19.06 ± 3.26	17.81 ± 3.49
Frustration	25.00 ± 4.33	24.06 ± 5.30	25.00 ± 5.27	18.75 ± 3.27	25.63 ± 5.19	15.94 ± 3.42	17.19 ± 2.74	14.06 ± 2.65

Table 6

The median for each takeover request type under the visual and auditory non-driving-related task conditions.

	Auditory icon	Earcon	Spearcon	Speech
Visual NDRT	4	2	3	1
Auditory NDRT	3.5	2.5	2.5	1.5
Overall	4	2	3	1

study were identical to those used in the study of Song et al. (2022), this discrepancy in results may be attributed to the difference between manual and automated driving. In automated vehicles, drivers are free from driving tasks and can engage in NDRTs. When TORs are issued, they must transfer their concentration from the NDRT to driving tasks. This transfer may cause drivers to struggle with grasping the meaning of auditory icons, although they are directly mapped to the driving-related event using drivers' prior knowledge. By contrast, in manually driven vehicles, drivers are constantly engaged in driving tasks. The above-mentioned cognitive transfer does not exist, so auditory icons can speed up drivers' responses via direct nomic mapping (Cabral and Remijn, 2019).

As for speech-based TORs, spercons, the compressed version of the speech, also did not result in faster takeover timing metrics than speech. This is consistent with the results of Jeon (2019). Nevertheless, spearcons were found to be able to shorten response times to a greater extent than speech warnings in the background of manual driving (Sabić et al., 2017). This discrepancy may also be attributed to the out-of-the-control-loop state of drivers during automated driving. Engaging in NDRTs may also have a detrimental effect on understanding sped-up speech due to cognitive distractions. Additionally, the spearcon TOR used in the present study was 40 % of the speech TOR in length, which was set in accordance with the previous studies (Šabić et al., 2021; Sabić and Chen, 2016). Although its effectiveness as a warning has been validated, it may have been compressed too much as a TOR to be understood easily. Hence, future studies are needed to consider the compression degree of speracons when exploring their effectiveness as TORs.

Taking the four types of auditory TORs together, the earcon might be most suitable for use as a TOR. The speech TOR was the most preferred by participants according to their subjective ranks, which is in line with the online survey studies of Bazilinsky and Winter (2015, 2018). However, the speech TOR led to longer takeover and lane change times than the earcon TOR (indicating poorer takeover response speeds), as well as a shorter min-TTC (indicating poorer takeover quality), so the speech TOR is less beneficial for driving safety. Contrastingly, the earcon TOR had the shortest takeover and lane change times, as well as the greatest min-TTC, among the four types of auditory TORs. Although its preference rank was lower than the speech TOR, it was still better than the auditory icon and spearcon TORs. Therefore, using earcons as TORs can aid in reaching a balance between subjective experience and driving safety. Moreover, takeover performance may be enhanced by combining various auditory TOR types. For example, adding a non-speech warning prior to the speech warning is a possible solution that can

simultaneously exploit their respective advantages. This approach has been proposed in the manual driving field (Sabić et al., 2021), and it deserves further study in the context of automated vehicles.

4.2. The impact of auditory non-driving-related tasks on takeover requests

Automated vehicles allow drivers to be free from driving tasks when the automated system is activated. Pfleging et al. (2016) revealed drivers' intention to engage in various NDRTs during automated driving. However, previous studies have mainly investigated the impact of NDRTs on takeover performance (Lee et al., 2021; Ma et al., 2020; Wandtner et al., 2018; see Naujoks et al., 2018 for a review). Although several studies have explored the influence of the NDRT modality on the effect of TORs (Petermeijer et al., 2017b; Yoon et al., 2019), they have mainly concentrated on the various modalities of TORs (e.g., all combinations of visual, auditory, and tactile modalities). Few studies have focused on the auditory TOR and considered the potential interference impact of auditory NDRTs on various types of auditory TORs. Therefore, we conducted a simulated driving experiment to further explore the moderating effect of the NDRT modality on the impact of various auditory TORs. It was found that when drivers are engaged in the visual NDRT, earcon TORs are more advantageous in terms of takeover time, lane change time, and min-TTC over the other types of auditory TORs. Nevertheless, these advantages were weakened when drivers were engaged in the auditory NDRT. The differences between the earcon TOR and the other TORs in most of the takeover metrics were no longer significant, except for the min-TTC. These results highlight the importance of considering the influence of auditory NDRTs when designing an auditory takeover interface.

Engaging in auditory NDRTs may decrease drivers' susceptibility to auditory TORs. According to multiple resources theory (Wickens, 2002), people's cognitive resources for each modality are limited, but people can simultaneously conduct dual tasks in separate modalities without mutual interference. Nevertheless, when the dual tasks occupy the same modality, they would interfere with each other. Hence, when engaging in auditory NDRTs, drivers' auditory cognitive resources, which can be devoted to the perception and processing of the TOR, are fewer than when they are engaging in visual NDRTs. This may account for the null differences among the four types of auditory TORs in the takeover timing metrics and workload under the auditory NDRT condition.

Notably, although the advantages of the earcon TOR in most metrics disappeared under the auditory NDRT condition, the earcon TOR also led to a greater min-TTC than the speech TOR, indicating a safer takeover maneuver. Therefore, using earcons as TORs also deserves recommendation to some extent when drivers are engaged in auditory NDRTs.

4.3. Implications for the design

Auditory interfaces are commonly used to present TORs (e.g., Ma et al., 2020; Petermeijer et al., 2017b; Wang et al., 2022; see Zhang et al., 2019a for a review). Designing an effective auditory TOR is, therefore, vital when aiming to ensure driving safety. The results of the present study revealed that the earcon TOR is generally the most

suitable among the four TOR types for alerting drivers to return to the control loop due to its advantages in the takeover timing and quality metrics. As for the auditory icon and spearcon, although they do have benefits as in-vehicle warnings in manually driven vehicles (Šabić et al., 2021; Song et al., 2022), they may not be appropriate for use as TORs in automated vehicles.

In addition, the findings of the present study highlight the importance for researchers and designers to consider the detrimental impact of auditory NDRTs when designing auditory takeover interfaces. The advantages of earcon TORs in the takeover timing metric would be weakened when drivers were engaged in auditory NDRTs. Related practitioners may further enhance auditory TORs by using a “pre-alert” (Ma et al., 2021; Wang et al., 2023) to forewarn drivers of the upcoming takeover operation or combining them with visual and tactile TORs to form multi-modal TORs (Petermeijer et al., 2017b; McDonald et al., 2019; Yoon et al., 2019).

4.4. Limitations

Although the present study was carefully prepared, there were also several limitations. First, the participants in this study were all undergraduate or graduate students of relatively young ages. As previous studies have shown, people’s auditory perception declines with increases in age (Tun et al., 2012), so it is necessary to expand the participant group and consider the role of hearing loss when conducting future studies. Second, within each type of auditory TOR, the present study only used one representative TOR chosen from previous literature. Although these TORs’ effectiveness or benefits have been demonstrated in previous studies, future studies should further explore the effects of various TOR types with more diverse candidates. Third, we did not record participants’ performance on the N-back tasks in the present study. Although participants were instructed to immerse themselves in the N-back task during the automated driving, it would have been useful to collect N-back task performance data to further confirm participants’ engagement in the task. Fourth, the N-back task used as an NDRT in the present study was an artificial task with relatively low ecological validity. Future studies are needed to explore the effect of NDRTs on auditory TORs through more ecological tasks, such as listening to music or holding a phone conversation.

5. Conclusion

The present study explored the effects of various auditory TORs (auditory icon, earcon, spearcon, speech) on takeover performance. By balancing driving performance and drivers’ preferences, we suggested the earcon TOR for alerting drivers to take over due to its advantages over the other TORs in takeover time, lane change time, and min-TTC, despite its suboptimal preferences ranking for drivers. Although the speech TOR was the most preferred by participants, it led to poorer takeover performance. Auditory icons and spearcons as TORs were also not found to be capable of improving takeover performance compared with earcons and speech, respectively. In addition, auditory NDRTs were found to have a detrimental impact on auditory TORs. That is, when drivers were engaged in the auditory NDRT, the advantages of earcon TORs in takeover time and lane change time disappeared. The results of this study highlight the importance of taking the interference impact of auditory NDRTs into consideration when designing auditory takeover interfaces in automated vehicles.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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